

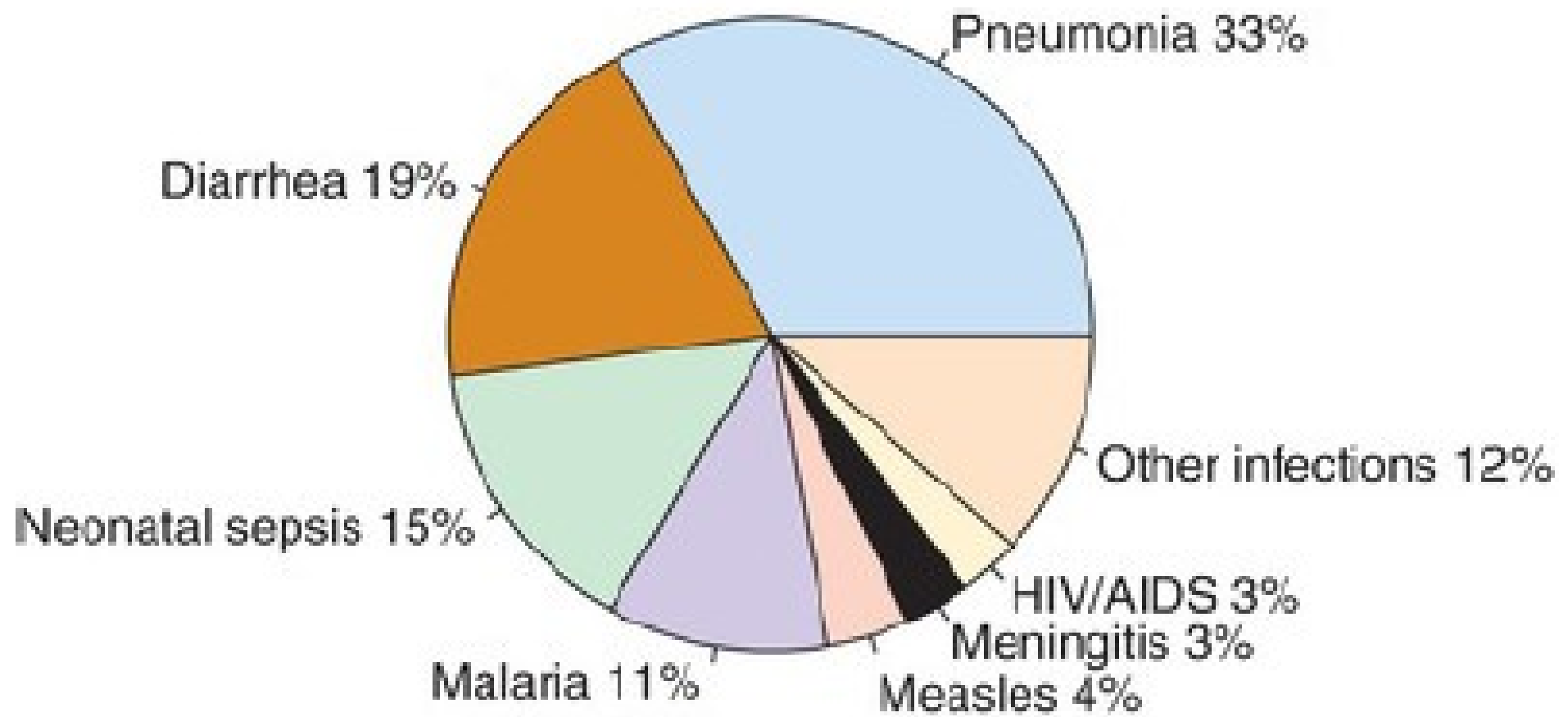
Lower respiratory tract infection

Pneumonia

- an inflammation of the parenchyma of the lungs

Epidemiology

- is the leading cause of death globally among children < 5 yr
- Accounting for an estimated 920,000 deaths each year
- **mortality is closely linked to poverty**
- > 99% of pneumonia deaths are in low- and middle-income countries
 - highest mortality rate occurring **in poorly developed countries in Africa and South Asia**



- In the US pneumonia mortality in children declined by 97%, probably as a result from
 - introduction of antibiotics
 - Vaccines
 - expansion of medical insurance coverage for children
- Vaccines which prevents from pneumonia
 - Haemophilus influenzae type b vaccine
 - Measles vaccine
 - Pertusis vaccine
 - Pneumococcal conjugate vaccines

Etiology

- The cause is difficult to determine because
 - **Direct culture** of lung tissue is **invasive and rarely performed**
 - Cultures performed on specimens obtained from the **upper respiratory tract or “sputum”** generally **do not accurately reflect the cause**

1. Infectious

- **Most common cause**
- Viral pathogens are a **prominent** cause in infants and children **<5 yr of age**
- Viruses can be detected in 40-80% of children with pneumonia using molecular diagnostic methods

- **RSV and rhinoviruses** - most commonly identified pathogens , especially in <2 yr of age
- Infection with **more than 1 respiratory virus** occurs in **up to 20% of cases**

- RSV is particularly **severe among infants and young children**
- Influenza virus causes disease and excess **hospitalization** for acute respiratory illness in **all age groups**
- Knowledge of the prevailing viral epidemic may lead to a presumptive initial diagnosis.

- **Immunization** status - fully immunized **against H. influenzae type b and S. pneumoniae** are less likely to be infected with these pathogens
- **immunosuppressed** or who have an **underlying illness** may be at risk for specific pathogens
 - Pseudomonas spp. in patients with cystic fibrosis

- **In Developed Country**

- S. pneumoniae is the most common bacterial pathogen
- Followed by C.pneumoniae and M. pneumoniae

- **In Developing Countries**

- S. pneumonia
- H.influenza
- S. aureus

- In **Immunocompromised** children
 - M. tuberculosis
 - atypical mycobacterium
 - Salmonella
 - E.coli
 - Pneumocystis jirovecii
 - cytomegalovirus

Table 400-3**Etiologic Agents Grouped by Age of the Patient**

AGE GROUP	FREQUENT PATHOGENS (IN ORDER OF FREQUENCY)
Neonates (<3 wk)	Group B streptococcus, <i>Escherichia coli</i> , other Gram-negative bacilli, <i>Streptococcus pneumoniae</i> , <i>Haemophilus influenzae</i> (type b,* nontypeable)
3 wk-3 mo	Respiratory syncytial virus, other respiratory viruses (rhinoviruses, parainfluenza viruses, influenza viruses, adenovirus), <i>S. pneumoniae</i> , <i>H. influenzae</i> (type b,* nontypeable); if patient is afebrile, consider <i>Chlamydia trachomatis</i>
4 mo-4 yr	Respiratory syncytial virus, other respiratory viruses (rhinoviruses, parainfluenza viruses, influenza viruses, adenovirus), <i>S. pneumoniae</i> , <i>H. influenzae</i> (type b,* nontypeable), <i>Mycoplasma pneumoniae</i> , group A streptococcus
≥5 yr	<i>M. pneumoniae</i> , <i>S. pneumoniae</i> , <i>Chlamydia pneumoniae</i> , <i>H. influenzae</i> (type b,* nontypeable), influenza viruses, adenovirus, other respiratory viruses, <i>Legionella pneumophila</i>

**H. influenzae* type b is uncommon with routine *H. influenzae* type b immunization.

Adapted from Kliegman RM, Marcandante KJ, Jenson HJ, et al: Nelson essentials of pediatrics, ed 5, Philadelphia, 2006, Elsevier, p. 504.

FUNGAL

Histoplasma capsulatum

Ohio/Mississippi River valley;
bird, bat contact

Blastomyces dermatitidis

Ohio/Mississippi River valley

Coccidioides immitis

Southwest United States

Cryptococcus neoformans

Bird contact

Aspergillus species

Immunosuppressed persons;
nodular lung infection

Mucormycosis

Immunosuppressed persons

Pneumocystis jiroveci

Immunosuppressed, steroids

RICKETTSIAL

Rickettsia rickettsiae

Tick bite

MYCOBACTERIAL

Mycobacterium tuberculosis

Travel to endemic region;
exposure to high-risk persons

Mycobacterium avium complex

Immunosuppressed persons

PARASITIC

Various parasites (e.g., *Ascaris*,
Strongyloides species)

Eosinophilic pneumonia

*Atypical pneumonia syndrome; may have extrapulmonary manifestations, low-grade fever, patchy diffuse infiltrates, poor response to β -lactam antibiotics, and negative sputum Gram stain.

From Kliegman RM, Greenbaum LA, Lye PS: Practical strategies in pediatric diagnosis & therapy, ed 2, Philadelphia, 2004, Elsevier, p. 29.

2. Noninfectious

- Aspiration of
 - Food or gastric acid
 - Foreign bodies
 - Hydrocarbons
 - Lipoid substances
- Hypersensitivity reactions
 - Drug induced
 - Radiation-induced pneumonitis

Pathogenesis

The lower respiratory tract possesses a number of defense mechanisms against infection

- mucociliary clearance
- macrophages
- secretory immunoglobulin A
- clearing of the airways by coughing

- it was believed that the LRT was—in the absence of infection—kept sterile
- **Currently, that the LRT contains diverse microbial communities**
- **disruption** of a complex lower respiratory ecosystem that is the site of **dynamic interactions** between **potential pneumonia pathogens**, **resident microbial communities**, and **host immune defenses**

Microorganisms gain access to the LRT in several ways

- aspiration from the oropharynx - most common
- Many pathogens are inhaled as contaminated droplets
- Rarely occurs via
 - Hematogenous spread
 - By contiguous extension from an infected pleural or mediastinal space

- **Viral pneumonia**

- Results from Spread of infection along the airways
- results direct injury of the respiratory epithelium, resulting in airway obstruction from swelling, abnormal secretions, and cellular debris.
- **Predispose to secondary bacterial infection by disturbing normal host defense mechanisms, modifying the bacterial flora**

- The **small caliber of airways in young infants** makes them particularly susceptible to **severe infection**
- Atelectasis, interstitial edema, and ventilation-perfusion mismatch causing significant hypoxemia often accompany airway obstruction

M. pneumoniae

- Attaches to the respiratory epithelium



- Inhibits ciliary action Cellular destruction and an inflammatory response in the submucosa.



- Air way obstruction causes by sloughed cellular debris, inflammatory cells, and mucus
- Spread of infection occurring along the bronchial tree, as it does in viral pneumonia.

S. pneumoniae

- Produces local edema that aids in the proliferation of organisms and their spread into adjacent portions of lung
- Often resulting in the characteristic **focal lobar involvement**

GAS

- Results in more diffuse infection with **interstitial pneumonia**
- The pathology includes
 - Necrosis of tracheobronchial mucosa
 - Formation of large amounts of exudate, edema, and local hemorrhage with extension into the interalveolar septa
 - Involvement of lymphatic vessels
- Increased likelihood of pleural involvement

***S. aureus* pneumonia**

- Manifests in confluent bronchopneumonia
- Is often unilateral
- Characterized by
 - Hemorrhagic necrosis
 - Irregular areas of cavitation of the lung parenchyma resulting in pneumatoceles
 - Empyema
 - bronchopulmonary fistulas -at times

Recurrent pneumonia

- *2 or more* episodes in a single year *or 3 or more* episodes ever, with **radiographic clearing between occurrence**
- An **underlying disorder** should be considered if a child experiences recurrent

Differential Diagnosis of Recurrent Pneumonia

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HEREDITARY DISORDERS

Cystic fibrosis
Sickle cell disease

DISORDERS OF IMMUNITY

HIV/AIDS

Bruton agammaglobulinemia
Selective immunoglobulin G subclass deficiencies
Common variable immunodeficiency syndrome
Severe combined immunodeficiency syndrome
Chronic granulomatous disease
Hyperimmunoglobulin E syndromes
Leukocyte adhesion defect

DISORDERS OF CILIA

Primary ciliary dyskinesia
Kartagener syndrome

ANATOMIC DISORDERS

Pulmonary sequestration
Lobar emphysema
Congenital cystic adenomatoid malformation
Gastroesophageal reflux
Foreign body

Tracheoesophageal fistula (H type)

Bronchiectasis

Aspiration (oropharyngeal incoordination)

Aberrant bronchus

CLINICAL MANIFESTATIONS

- Preceded by several days of symptoms of URTI typically rhinitis and cough
- In viral pneumonia, fever is usually present but temperatures are **generally lower than in bacterial** pneumonia

Bacterial pneumonia In older children and adolescents

- Brief URTI is followed by the **abrupt onset** of shaking chills and high fever
- Accompanied by drowsiness with intermittent periods of restlessness; rapid respirations; a dry, hacking, unproductive cough; anxiety; and, occasionally, delirium

Circumoral cyanosis may be observed

- Many children are noted to be splinting on the affected side to minimize pleuritic pain and improve ventilation
- They may lie on their side with their knees drawn up to their chest

Physical examination

- **V/S**
 - **Tachypnea is the most consistent clinical manifestation of pneumonia**
- **G/A**
 - **Increased work of breathing accompanied by intercostal, subcostal, and suprasternal retractions, nasal flaring, and use of accessory muscles is common**
- **Severe infection may be accompanied by cyanosis and respiratory fatigue, especially in infants**

- **Chest**

- Findings depends on the stage of pneumonia
- Early in the course of illness, diminished breath sounds, scattered crackles, and rhonchi are commonly heard over the affected lung field.
- With the development of increasing consolidation or complications such as effusion, empyema, or pyopneumothorax, dullness on percussion is noted and breath sounds may be diminished.
- A lag in respiratory excursion often occurs on the affected side.

- **Abdomen**

- Abdominal distention because of gastric dilation from swallowed air or ileus.
- Abdominal pain is common in lower lobe pneumonia.
- The liver may seem enlarged because of downward displacement of the diaphragm secondary to hyperinflation of the lungs or superimposed congestive heart failure.
- Nuchal rigidity, in the absence of meningitis, may also be prominent, especially with involvement of the right upper lobe

Bacterial pneumonia In infants

- **There may be a prodrome of upper respiratory tract infection and diminished appetite, leading to the abrupt onset of fever, restlessness, apprehension, and respiratory distress.**
- **These infants appear ill with respiratory and often cyanosis.**
- **Results of physical examination may be misleading with meager findings disproportionate to the degree of tachypnea.**

- **Some infants may have associated GI disturbances characterized by vomiting, anorexia, diarrhea, and abdominal distention secondary to a paralytic ileus.**
- **Rapid progression of symptoms is characteristic in the most severe cases of bacterial pneumonia.**

Diagnosis

- **CBC**
- **CXR**
 - **Confirms the diagnosis**
 - **Indicate a complication**
 - **Repeat chest x-rays are not required for proof of cure for patients with uncomplicated pneumonia**

- An infiltrate on chest radiograph (PA and lateral views) supports the diagnosis of pneumonia
- Viral pneumonia - hyperinflation with bilateral interstitial infiltrates and peribronchial cuffing
- Confluent lobar consolidation - typically seen with pneumococcal pneumonia
- The radiographic appearance **alone is not diagnostic**
 - other clinical features must be considered

- **current guidelines** do not recommend that a CXR for suspected pneumonia who are **well enough to be managed as outpatients**
 - only rarely changes management
- Chest ultrasonography - highly sensitive and specific
 - determining lung consolidations and air bronchograms or effusions

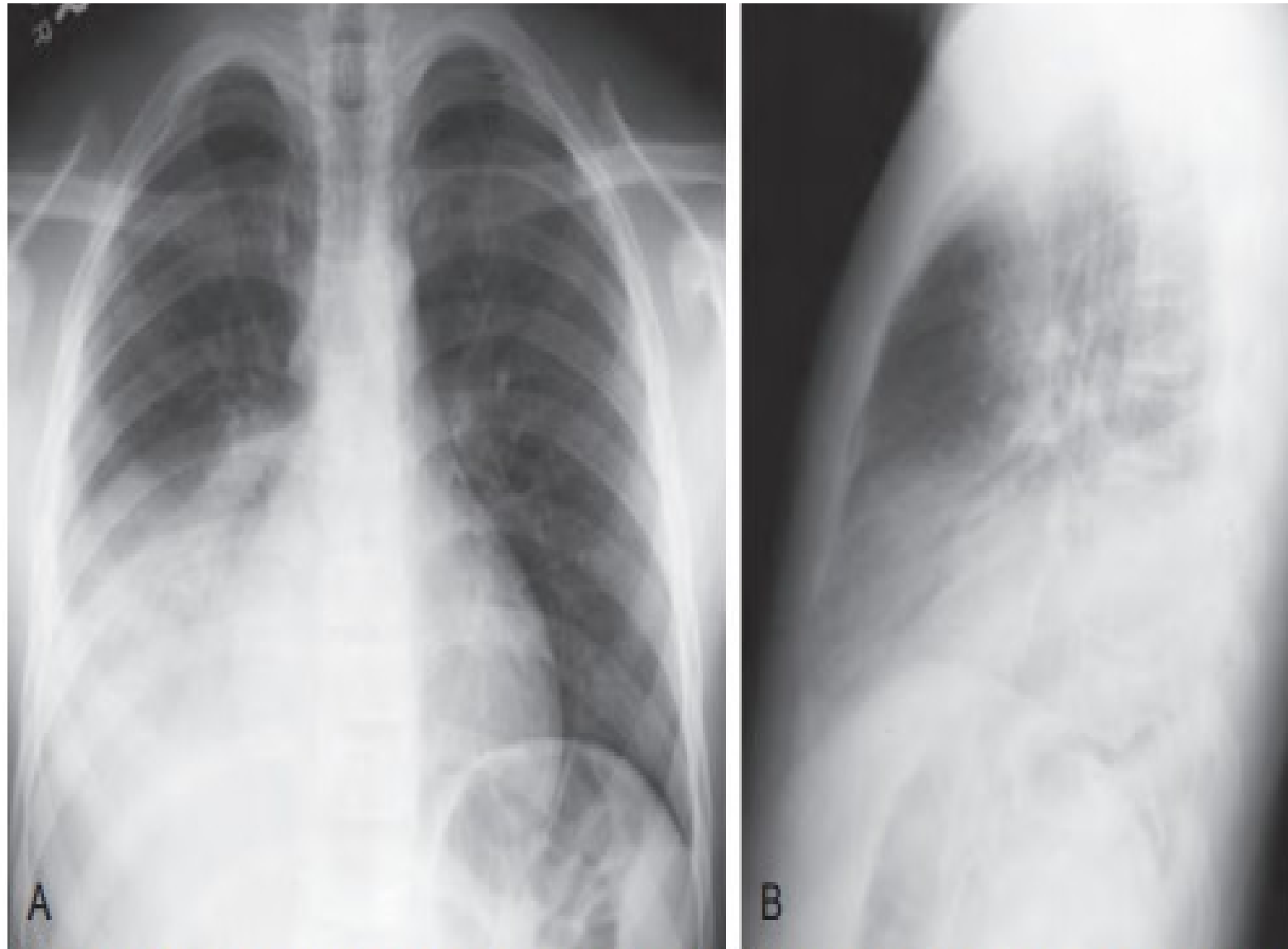


Figure 400-3 Radiographic findings characteristic of pneumococcal pneumonia in a 14 yr old boy with cough and fever. Posteroanterior (A) and lateral (B) chest radiographs reveal consolidation in the right lower lobe, strongly suggesting bacterial pneumonia.

Isolation of organisms

Culture

- The definitive diagnosis of a bacterial infection requires isolation of an organism from the **blood, pleural fluid, or lung**
- **Culture of sputum** - little value in the diagnosis of pneumonia in young children
- Blood cultures are positive in only 10% of children with pneumococcal pneumonia
- Growth of respiratory viruses in tissue culture usually requires 5–10 days

- Blood cultures are recommended for those who
 - Fail to improve
 - clinical deterioration
 - With complicated pneumonia
 - Requires hospitalization

PCR

- Reliable DNA or RNA tests for the rapid detection of RSV, parainfluenza, influenza, and adenoviruses are available and accurate.

Serologic

- Can also be used to diagnose a recent respiratory viral infection
- Require testing of acute and convalescent serum samples for a rise in antibodies to a specific viral agent.

- With the use of molecular diagnostic testing, a bacterial or viral cause of pneumonia can be identified in 40-80%

bacterial and viral pneumonia

D/C	Bacterial	Viral
presentation	HGF	
WBC	15,000-40,000/mm³	normal or elevated but not higher than 20,000/mm³
differential	granulocyte	lymphocyte
CXR	A large pleural effusion, lobar consolidation	

Treatment

- Treatment of suspected bacterial pneumonia is based on the presumptive cause and the clinical appearance of the child

A. Out patient management

- For mildly ill children who do not require hospitalization
 - Amoxicillin is recommended.
 - In communities with a high percentage of penicillin-resistant pneumococci, high doses of amoxicillin (80–90 mg/kg/24 hr) should be prescribed.
 - Therapeutic alternatives include cefuroxime axetil or amoxicillin/clavulanate.

- For atypical pneumonias
 - For school-aged children Macrolide such as azithromycin is an appropriate choice
 - In adolescents, a respiratory fluoroquinolone (levofloxacin, gatifloxacin, moxifloxacin, gemifloxacin) may be considered

Factors Suggesting Need for Hospitalization

- Age <6 mo
- Immunocompromised state
- Toxic appearance
- Moderate to severe respiratory distress
- Hypoxemia (oxygen saturation <90% breathing room air, sea level)
- Complicated pneumonia*
- Sick cell anemia with acute chest syndrome
- Vomiting or inability to tolerate oral fluids or medications
- Severe dehydration
- No response to appropriate oral antibiotic therapy
- Social factors (e.g., inability of caregivers to administer medications at home or follow-up appropriately)
- * Pleural effusion, empyema, abscess, bronchopleural fistula, necrotizing pneumonia
- acute respiratory distress syndrome, extrapulmonary infection (meningitis, arthritis, pericarditis, osteomyelitis, endocarditis), hemolytic uremic syndrome, or sepsis

B. Hospitalized patients

- The empirical treatment of suspected bacterial pneumonia requires an approach based on the clinical manifestations at the time of presentation

- Ampicillin or penicillin G indicated
 - In areas without substantial high-level penicillin resistance among *S. pneumoniae*
 - Children who are fully immunized against *H. influenzae* type b and *S. pneumoniae*
 - Children who are not severely ill
- For children who do not meet these criteria, ceftriaxone or cefotaxime should be used.

- If clinical features suggest staphylococcal pneumonia (pneumatocelles, empyema), initial antimicrobial therapy should also include vancomycin or clindamycin.

- **Up to 30% of patients** with known viral infection may have **coexisting bacterial pathogens**
- **deterioration in clinical status should signal the possibility of superimposed bacterial infection and antibiotic therapy should be initiated**

Duration of treatment

- continued until the patient has been **afebrile for 72 hr**
- The total duration **should not be less than 10 days (or 5 days if azithromycin is used)**
- Shorter courses (5-7 days) may also be effective, particularly for children managed on an outpatient basis, but further study is needed

- Oral zinc may **reduces mortality** in severe pneumonia - developing countries
- supportive care and may require intravenous fluid
- respiratory support (INO2,CPAP , MV)
- **Bubble CPAP improves mortality compared with standard oxygen therapy in settings without access to ventilator derived CPAP or mechanical ventilation**
- vasoactive medications for hypotension or sepsis physiology

Response To Treatment

- uncomplicated community-acquired bacterial pneumonia respond to therapy with improvement in
 - **Clinical symptoms** (fever, cough, tachypnea, chest pain) within 48–96 hr of initiation of antibiotics
 - **Cough** 6-10days
 - **Radiographic evidence** of improvement substantially lags behind clinical improvement

- A number of possibilities must be considered when a patient does not improve with appropriate antibiotic therapy
- **A repeat chest x-ray is the 1st step in determining the reason for delay in response to treatment**
- Broncho alveolar lavage - in children with **respiratory failure**
- High resolution CT scans -identify **c omplications or an anatomic reason** for a poor response

DDX

1 complications

2 bacterial resistance

3 nonbacterial etiologies such as viruses and aspiration of FB or food

4 bronchial obstruction from endobronchial lesions, foreign body, or mucous plugs;

5 pre-existing diseases

6 other noninfectious causes (including bronchiolitis obliterans, hypersensitivity pneumonitis, eosinophilic pneumonia, aspiration, and Wegener granulomatosis)

Complications

- usually the result of
 - Direct spread of bacterial infection within the thoracic cavity
 - Pleural effusion
 - Empyema
 - Pericarditis
 - Bacteremia and hematologic spread
 - Meningitis
 - Suppurative arthritis
 - Osteomyelitis

Empyema

- *S. aureus*, *S. pneumoniae*, and *S. pyogenes* are the most common causes of parapneumonic effusions and of empyema
- ultrasonography and CT
 - helpful in determining the stage of empyema

- The treatment of empyema is based on the stage

- Exudative

- Fibrinopurulent

- Organizing

3 stages of empyema

Stage 1: Exudative → early stage

Think: "Fluid stage."

Stage 2: Fibrinopurulent.

Fluid → bacteria + pus + fibrin

Stage 3: Organizing → late stage

The pleura becomes thick and scarred

Empyema = antibiotics + remove the pus

- The mainstays of therapy
 - **antibiotic therapy**
 - **drainage with tube thoracostomy** with instillation of fibrinolytic therapy (urokinase, streptokinase, alteplase)
- Selected video-assisted thoracoscopy (**VATS**) to debride, lyse adhesions, and drain loculated areas of pus
- **Early diagnosis and intervention**, particularly with VATS, may obviate the **need for thoracotomy and open debridement**.

Features Differentiating Exudative From Transudative Pleural Effusion

FEATURE	TRANSUDATE	EXUDATE
Appearance	Serous	Cloudy
Leukocyte count	<10,000/mm ³	>50,000/mm ³
pH	>7.2	<7.2
Protein	<3.0 g/dL	>3.0 g/dL
Ratio of pleural fluid protein to serum	<0.5	>0.5
LDH	<200 IU/L	>200 IU/L
Ratio of pleural fluid LDH to serum	<0.6	>0.6
Glucose	≥60 mg/dL	<60 mg/dL

Prognosis

- Mortality from community-acquired pneumonia in developed nations is rare
- Most children with pneumonia do not experience long-term pulmonary sequelae

Prevention

- Vaccination against
 - Pneumococcal (PCV—13) in US
 - Influenza vaccine recommendations to include all children >6 mo of age

<2 years → b/c bronchioles are very narrow
peak → 2–6 months
usually seasonal
Transmission
Respiratory droplets
Direct contact
Contaminated hands/surfaces
Diagnosis → clinical diagnosis
Young child + viral prodrome + first episode of
wheezing/crackles + increased respiratory effort

Pleurisy = inflammation of the pleura
→ sharp
→ worse with deep inspiration
→ worse with coughing

mgt
→ Treat the underlying cause
→ Pain control

Read on

Small airway + small amount of swelling → major obstruction
no routine antibiotic treatment because bronchiolitis is usually viral

- Bronchiolitis
- Pleurisy, pleural effusion and empyema
- Bronchiectasis
- Pulmonary abscess

Lung abscess = pus INSIDE the lung

🌀 Bronchiectasis = permanent abnormal dilation and structural damage of the bronchi due to chronic infection/inflammation.
or Permanent airway dilation
🕒 occur at any age

C/F
Chronic cough—productive/wet
Sputum
purulent
foul smelling
Crackles
Recurrent pneumonia
Hemoptysis
Clubbing
Poor growth

DX
CXR
increased bronchovascular markings
tram-track appearance
ring shadows
HRCT chest
Bronchial dilation
Signet-ring sign (ring = bronchus, dot = artery)
Lack of normal tapering
Bronchial wall thickening
Tree-in-bud changes

Thank you